Hawkes Process Estimation on High-Frequency Cryptocurrency Data via Neural Networks

Trabalho de Graduação — Engenharia de Computação, ITA

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Motivation

- Price changes cluster: a move tends to be followed by more moves.
- The Hawkes process models this self-excitation with few, interpretable parameters.
- Estimating it by maximum likelihood (MLE) needs an optimization per window.
- **Amortized estimation**: train a network once on simulated paths; each new window then costs a single forward pass (Lee, 2023).
- Lee (2023): an LSTM on U.S. equities (NBBO data, not public).

This work: reproduce Lee on public crypto data, test attention, and take the estimators to real windows.

Symmetric bivariate Hawkes process

Up-moves N_1 and down-moves N_2, exponential kernel:

$$\lambda_1(t) = \mu + \sum_{t_k^1 < t} \alpha_{\text{self}} e^{-\beta(t - t_k^1)} + \sum_{t_k^2 < t} \alpha_{\text{cross}} e^{-\beta(t - t_k^2)}$$

and symmetrically for λ_2.

- μ: exogenous activity
- α_{self}: persistence of direction
- α_{cross}: reversion
- β: speed of forgetting

Stability (branching ratio):

$$g = \frac{\alpha_{\text{self}} + \alpha_{\text{cross}}}{\beta} < 1$$

$g \to 1$: long cascades, near-critical.

Research question and hypotheses

Question

Can an attention-based architecture estimate the parameters of a 2D Hawkes process on crypto price moves with accuracy and stability equal to or better than Lee's LSTM, while keeping inference faster than the MLE?

Registered before the final run:

- **H1.** Binance's public USD-M data yield sequences structurally equivalent to Lee's.
- **H2.** The LSTM reaches MLE-level error with substantially lower latency.
- **H3.** Attention improves error or robustness on long-range dependencies.

Data: Binance USD-M perpetual futures

- BTCUSDT and ETHUSDT, `bookTicker` (best bid/ask)
- SHA-256 manifest, audited schema
- Mid-price on a 100 ms grid (last state per bin)
- Event = mid-price change (up / down)
- Chronological splits, no window crosses a split

Window	Events	Windows
Primary (Jan. 2024)	1,826,904	1,813
Robustness (Oct. 2023)	1,288,352	1,281

Calibration block (15–22 Jan.):

- 76% of moves repeat direction
- median jump 10–15 ticks
- 37–39% of gaps equal the grid step

Simulation: Ogata thinning, empty initial history.

Constrained MLE

$$(\mu, \alpha_{\text{self}}, \alpha_{\text{cross}}, d) = \text{softplus}(z)$$
$$\beta = \alpha_{\text{self}} + \alpha_{\text{cross}} + d$$

L-BFGS-B, analytic gradient, 5 heuristic starts; never initialized at the truth.

Networks (input per event: $(\log(1 + \Delta t), \pm 1)$; same output head, $g < 1$ by construction)

	LSTM	Transformer
Layers	2 LSTM×12	2 enc., d=32
Heads / FF	–	4 / 128
Parameters	2,068	25,700
Effective batch	64	8
Best epochs	164–299	38–73
Training (5 seeds)	41 h	110 h

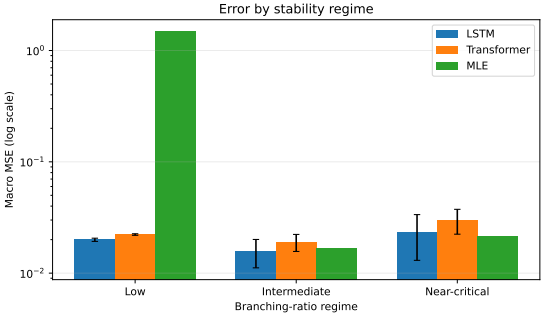
Frozen synthetic protocol

- 75,000 / 10,000 / 15,000 paths, all of 1,000 events
- Prior: $\mu \in [0.2, 0.6]$, $\beta \in [0.8, 2.0]$, self-excited fraction $\in [0.2, 0.8]$
- g balanced across three strata: low $[0.10, 0.35)$, intermediate $[0.35, 0.70)$, near-critical $[0.70, 0.95]$
- Loss: MSE on the original scale; Adam 10^{-3}; 300 epochs, patience 30
- 5 seeds per network; MLE fitted on every test path
- Test set opened once; tables and figures generated from the artifacts

Synthetic results

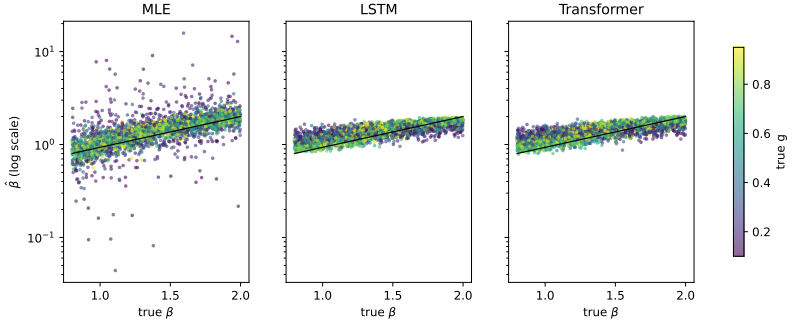
Estimator	Macro MSE	MSE of g
LSTM	0.0196	0.0018
Transformer	0.0237	0.0021
MLE	0.506	0.0023

- LSTM vs Transformer: not significant across seeds (Welch $p = 0.19$)
- LSTM ensemble better on 70% of paths, in every g regime



Why the MLE "loses": identifiability of β

- 94% of the MLE's squared error in β comes from 1% of the paths, almost all with low g
- In all 300 worst paths the likelihood at the estimate exceeds the likelihood at the truth
- Networks shrink toward the prior and never leave $\beta \in [0.8, 2.0]$
- Typical path: median loss MLE 0.149 vs LSTM 0.151



Lee's fixed point and computational cost

Fixed point $(0.3, 0.4, 0.7, 1.5)$, 2,000 paths

- MLE nearly unbiased; SD $\approx \sqrt{2}\times$ Lee's (half the length)
- Networks shrink $\alpha_{\text{self}}, \alpha_{\text{cross}}$ toward the prior, as in the paper
- Macro MSE: MLE 0.0103, LSTM 0.0111
- Heuristic and oracle MLE give identical estimates

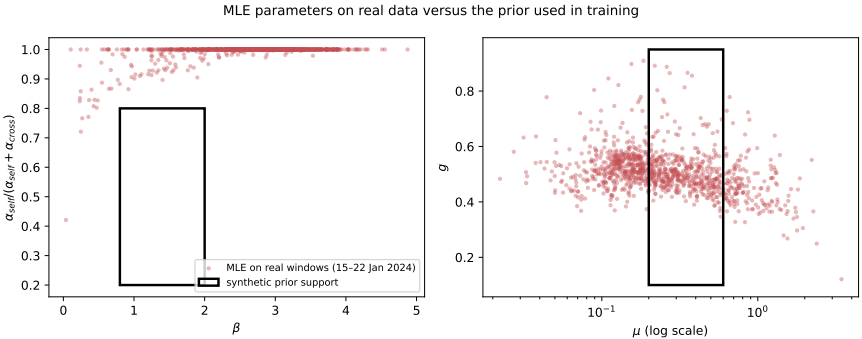
Cost per path (controlled benchmark)

MLE, 5 starts, one core	9.6 ms
LSTM, batch 1, CPU	7.0 ms
Transformer, batch 1, CPU	4.8–9.9 ms
LSTM, batch 256, MPS	0.07 ms
Lee: network / MLE	12 / 1,763 ms

Lee's $\sim$150$\times$ gain is not reproduced at batch size one.

Real windows lie outside the training prior

MLE on 1,063 calibration windows: median $\beta = 2.71$ (prior: ≤ 2.0); $\alpha_{\text{cross}} < 10^{-3}$ in 89% of windows; 0% of windows inside all prior ranges.

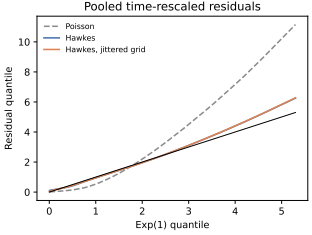


Networks trained on the synthetic prior cannot reach this region.

Goodness of fit on real data

Data	KS rej.	LB rej.
Hawkes, real, jittered	70.3%	47.1%
Simulated on grid	47.7%	4.7%
Simulated, continuous	0.9%	4.9%

- Hawkes $\gg$ Poisson: LR rejects Poisson in 99.8%; median ΔAIC -721
- The grid alone inflates KS
- Ljung–Box keeps its size on the grid, but rejects 47% on real data
- Same pattern on the test block (274 windows): LR 100%, Ljung–Box 47% vs 4%





Target on real data = MLE of an approximating model.

Networks on real windows (scope A)

Label = MLE of each window. Train 15–22 Jan. (1,063), select 23–25 Jan. (476), test 26–28 Jan. (274), opened once. 3.0 h of machine time.

Variant (5-seed ensemble)	Validation	R^2_β (val.)	Test	October
Training-label mean	0.210	0.00	0.218	0.629
LSTM, synthetic weights	0.555	−1.19	0.250	1.115
Transformer, synthetic weights	0.558	−1.08	0.275	1.060
LSTM, from scratch	0.139	0.30	0.250	0.409
Transformer, from scratch	0.100	0.47	0.169	0.333
LSTM, fine-tuned (selected)	0.036	0.82	0.044	0.121
Transformer, fine-tuned	0.041	0.79	0.040	0.083

Macro MSE against the MLE label. October 2023: never used for training or selection.

Fine-tuned LSTM vs Transformer (post hoc)

Paired by window; loss = squared error averaged over the 4 parameters; difference = LSTM − Transformer.

Block	Difference	Bootstrap 95% CI	Wilcoxon p
Test (274 windows)	0.0046	$[-0.0020,\ 0.0112]$	0.385
October 2023 (1,281)	0.0384	$[0.0319,\ 0.0449]$	$< 10^{-32}$

- Test: the two variants cannot be told apart
- October: the Transformer has the lower loss in 64% of windows
- Not pre-registered: the selection had already been made on validation

Only the fine-tuned networks are usable; the Transformer degrades less under a change of regime.

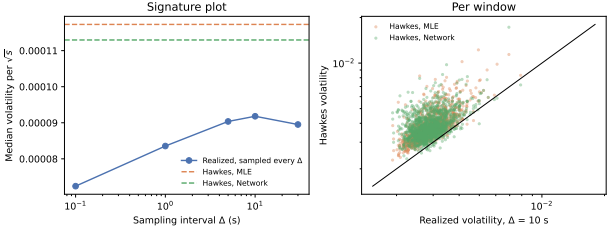
Hawkes volatility

$$\sigma_H^2 = 2\Lambda\left[\mathrm{Var}(J) + \frac{\mathbb{E}[J]^2}{(1-a_s+a_c)^2}\right], \quad \Lambda = \frac{\mu}{1-a_s-a_c}$$

J = absolute log return per event; compared with the realized volatility of the log mid-price sampled every $\Delta = 10$ s.

Level about right (ratio ≈ 1.25); the network ranks windows worse. Addendum (Feb 2024): Transformer 0.69, LSTM 0.47, MLE 0.77.

Block	Params	Ratio	ρ_S
Validation	MLE	1.24	0.67
Validation	network	1.24	0.47
Test	MLE	1.27	0.78
Test	network	1.38	0.28



Addendum (ADR 009): a new week, read once

Same data and seeds; one candidate chosen on validation, then compared with the frozen LSTM on 5–11 Feb 2024 (711 windows, never used before).

Candidate	Validation	New week	October
LSTM, frozen (selected in the TG)	0.0359	0.0264	0.121
Transformer, frozen	0.0413	0.0285	0.083
LSTM, 1,000-epoch cap (selected)	**0.0294**	0.0212	0.116
LSTM, warm-up + plateau decay	0.0358	0.0260	0.131
LSTM, standardized loss	0.0354	0.0315	0.102
Average LSTM + Transformer	0.0304	**0.0204**	0.087

- Primary: -0.0052, CI $[-0.0060, -0.0044]$, lower loss in 73% of windows
- The average also wins (Holm $p < 10^{-12}$); schedule ties; standardized loss loses

The LSTM was undertrained; the average of the two networks has the lowest loss on the new week.

Addendum: how much of β the 100 ms grid hides

Simulate in continuous time, take to the grid (one event per cell), fit the MLE; invert the median map for each real window.

- Grid label $\approx$ 58% of the continuous β
- January: 2.71 on the grid $\leftrightarrow$ 4.7 continuous
- October: 3.19 $\leftrightarrow$ 6.5
- Mean memory $\approx$ 210 ms, not 370 ms
- Jitter breaks ties, cannot restore merged events

True β	$\mu = 0.05$	0.23	1.0
1.2	1.04	0.93	0.63
2.7	2.11	1.88	1.26
4.5	3.17	2.72	1.83
8.0	4.63	4.01	2.65
13.0	5.97	5.02	3.25

Median MLE β on the grid

The labels describe the process seen through the grid, not in continuous time.

Addendum (ADR 010): a prior built from the real labels

Reduced scope B: invert January labels to continuous time (+ noise, 25% wide box), simulate, take to the grid, label by the MLE; pretrain the LSTM, fine-tune (1,000-epoch recipe). Second new week, read once: 4–10 Mar 2024 (1,860 windows).

Candidate	Validation	March	February	October
LSTM, 1,000-epoch cap (ADR 009)	0.0294	0.109	0.0212	0.116
Average LSTM + Transformer (ADR 009)	0.0304	0.102	0.0204	0.087
Scope B LSTM, no fine-tuning	0.1439	0.122	0.2115	0.141
Scope B LSTM, fine-tuned	0.0202	**0.040**	0.0153	**0.034**
Scope B LSTM + Transformer	**0.0196**	0.056	**0.0137**	0.041

- Primary: -0.068, CI $[-0.082, -0.056]$: 63% lower loss, wins in 72% of windows
- Gain is in μ (March MSE 0.31 $\rightarrow$ 0.086): extrapolates to higher activity

A prior built from the real labels helps more than a longer budget; it still needs fine-tuning.

Addendum (ADR 011): can a network undo the grid?

LSTM trained on gridded scope B paths with the continuous-time truth as target; 2,000 new paths of the same prior.

Estimator	Macro MSE	Bias β	|Error| β
LSTM, trained on the truth	**0.33**	$+2.8\%$	8.9%
MLE on the grid	3.95	-44.1%	44.1%
MLE corrected by the map	0.65	-2.4%	**8.4%**

- Primary (network vs corrected MLE): -0.33, CI $[-0.40, -0.26]$; map better on the typical path
- Real windows: network $\beta \approx 7.7$, map 4.9, grid 2.76 (Spearman 0.55): they disagree

The correction works in simulation but cannot yet be trusted on real windows.

Assessment of the hypotheses

- H1: supported as to structure and volume; but persistent direction, multi-tick jumps and grid-discretized times.
- H2: partially supported. LSTM error of the MLE's order; lower latency only in batched inference, not against an efficient MLE at batch size one.
- H3: not supported on synthetic data: attention did not reduce error, not even in the near-critical regime.
- Real windows: the fine-tuned networks tie on the test block; the fine-tuned Transformer degrades less in October (post hoc); on a pre-registered new week the LSTM + Transformer average is best: attention complements the LSTM; with a prior built from the real labels, the LSTM alone is best.

Limitations

- No hyperparameter search; post-LN Transformer without warm-up; batch 8 vs 64
- Five seeds; the fine-tuned LSTM stopped near the 300-epoch cap (addendum: a longer budget helps)
- On real data the label is the MLE of a misspecified model; α_{cross} on the boundary of the parameter space; labels biased by the grid
- Synthetic metrics depend on a prior not calibrated on the data (scope B: LSTM only, one prior design)
- Two assets, two windows; latency depends on hardware and batch
- Lee's paper omits ranges, seeds and training details

Future work

- Extend the prior built from the real labels to the Transformer and more weeks
- Make the grid correction trustworthy on real data (grid-aware likelihood, prior covering the misspecification)
- Richer kernels to absorb the residual dependence (sums of exponentials, power law)
- Transformer: pre-LN, continuous-time encodings, hyperparameter search (warm-up and standardized targets did not help in the addendum)
- Use the estimators for volatility forecasting, not only description

Conclusion

- Reproducible pipeline from public Binance data to a frozen comparison of three estimators; Lee's qualitative pattern reproduced at the fixed point
- Attention brought no accuracy gain on synthetic data
- The networks' aggregate-MSE advantage comes from robustness where β is poorly identified, and from shrinkage toward the prior
- Crypto occupies another region of the parameter space; the networks become usable on real windows only with synthetic pretraining plus fine-tuning on MLE labels
- Hawkes volatility gets the level right; the MLE ranks windows best
- Addendum: longer training and an LSTM + Transformer average beat the selected LSTM on a new week; a prior built from the real labels cuts the loss by 63% on another

Main benefit of amortized estimation: robustness and batched cost, not per-path accuracy.

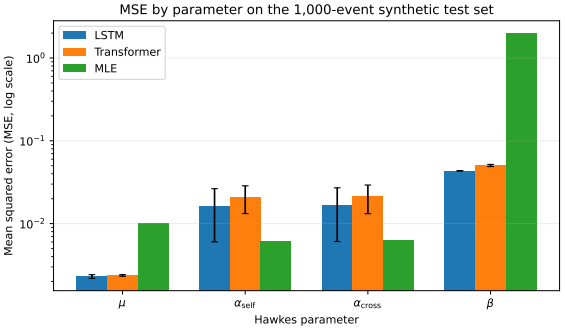
Thank you

Code, artifacts and protocol: private repository, release `resultados-finais`

Backup: test MSE by parameter (synthetic)

	μ	α_{self}	α_{cross}	β
LSTM	0.0023	0.016	0.017	0.043
Transf.	0.0024	0.021	0.021	0.050
MLE	0.0101	0.006	0.006	2.002

MLE is best for α_{self} and α_{cross}; β carries 98.9% of its squared error.



Backup: Lee's fixed point (mean / SD over 2,000 paths)

	$\mu = 0.3$		$\alpha_{\text{self}} = 0.4$		$\alpha_{\text{cross}} = 0.7$		$\beta = 1.5$	
Estimator	Mean	SD	Mean	SD	Mean	SD	Mean	SD
LSTM (ensemble)	0.304	0.037	0.495	0.054	0.625	0.061	1.532	0.144
Transformer (ensemble)	0.296	0.039	0.528	0.056	0.606	0.062	1.531	0.154
MLE (heuristic = oracle)	0.307	0.037	0.396	0.073	0.708	0.086	1.516	0.164
Lee (2023) network	0.315	0.036	0.443	0.066	0.573	0.070	1.574	0.136
Lee (2023) MLE	0.304	0.031	0.399	0.050	0.702	0.061	1.508	0.115

Lee: 100,000 paths of 2,000 steps; here 2,000 paths of 1,000 events.

Backup: October 2023 by volatility tercile

Variant	All	Low vol.	Medium vol.	High vol.
Training-label mean	0.629	0.500	0.333	1.055
LSTM, from scratch	0.409	0.405	0.329	0.492
Transformer, from scratch	0.333	0.449	0.312	0.238
LSTM, fine-tuned (selected)	0.121	0.108	0.071	0.185
Transformer, fine-tuned	0.083	0.034	0.039	0.176

1,281 windows, MLE converged in all; weights stored before the test was opened.

Backup: Hawkes volatility signature (test block)

